Week 13

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the standard error and conservative degrees of freedom for a difference in two independent means. (CLO 5) **2.** Check the conditions for two-sample t procedures. (CLO 5) **3.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$. (CLO 6) **4.** Explain how the order of the groups affects the sign of the difference. (CLO 6) **5.** Perform a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4-CI Confidence Interval for a Difference in Means e-Text: 6.4-D Distribution of a Difference in Means e-Text: 6.4-HT Hypothesis Test for a Difference in Means Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test Your Turn: Exercise and Sleep (two-sample t test and CI)	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
6. Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8) **7.** Define the difference variable d and the order of subtraction. (CLO 8) **8.** Perform a paired t test. (CLO 8) **9.** Construct and interpret a paired t confidence interval. (CLO 6) **10.** Explain why pairing reduces variability. (CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – define d and the order of subtraction Your Turn: Paired or Unpaired? (class discussion of study scenarios)	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM) Take-Home Quiz 8 (due Mon Nov 23, 11:59 PM) – covers 6.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus